TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	14.7 ± 29.40
Correlation Stability	37.0 ± 21.14
Distribution Consistency	14.5 ± 29.01
Entropy Preservation	14.4 ± 20.57
Mean Z Score Drift	0.2 ± 0.26
Categorical Fidelity	17.4 ± 14.21
Privacy Risk	6.6 ± 10.15
Data Density	40.0 ± 48.99
Coverage Score	25.2 ± 33.93
Feature Accuracy	26.9 ± 33.58
Novelty Score	0.2 ± 0.27

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Dataset Audit: breast_cancer_coimbra.csv

1. Dataset Portrait

Attribute	Value
Rows	116
Columns	10
Numeric features	10
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	14.7 ± 29.40
Correlation Stability (*2)	37.0 ± 21.14
Distribution Consistency (*3)	14.5 ± 29.01
Entropy Preservation (*4)	14.4 ± 20.57
Mean Z Score Drift (*5)	0.2 ± 0.26
Categorical Fidelity (*6)	17.4 ± 14.21
Privacy Risk (*7)	6.6 ± 10.15
Data Density (*8)	40.0 ± 48.99
Coverage Score (*9)	25.2 ± 33.93
Feature Accuracy (*10)	26.9 ± 33.58
Novelty Score (*11)	0.2 ± 0.27

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4958333333333333	N/A	0.4916666666666667	-0.00416666666666674	
F1	0.6547289527475286	N/A	0.6336079485145698	-0.02112100423295869	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7458333333333333	N/A	0.7416666666666666	-0.00416666666666652	
F1	0.754839950419102	N/A	0.7453161408952925	-0.009523809523809523	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7	N/A	0.7208333333333333	0.020833333333333336	
F1	0.7268772302965851	N/A	0.7382196573910128	0.011342427094427732	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4625	N/A	0.4875	0.025000000000000005	
F1	0.5361601301923884	N/A	0.5056830517153099	-0.030477078477078497	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.575	N/A	0.5916666666666666	0.016666666666666667	
F1	0.6619575575457929	N/A	0.5898506612878137	-0.07210689625797917	

Model: DT

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7041666666666666	N/A	0.7625	0.05833333333333335	
F1	0.7246672490150751	N/A	0.7700806281840765	0.04541337916900134	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7583333333333334	N/A	0.7541666666666667	-0.00416666666666674	
F1	0.7779203016013361	N/A	0.77112418217853	-0.006796119422806069	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6416666666666667	N/A	0.6416666666666667	0	
F1	0.5557236965932618	N/A	0.5557236965932618	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7083333333333334	N/A	0.7	-0.00833333333333337	
F1	0.7179811943844927	N/A	0.7056240663866326	-0.012357127997860262	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.725	N/A	0.725	-1.1102230246251566e-17	
F1	0.7359528619528619	N/A	0.7321686429512516	-0.003784219001610312	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	48.0	23.5	70.0	2.707	0.467408667	8.8071	9.7024	7.99585	417.114	0.0
2	83.0	20.69049454	92.0	3.115	0.706897333	8.8438	5.429285	4.06405	468.786	0.0
3	82.0	23.12467037	91.0	4.498	1.009651067	17.9393	22.43204	9.27715	554.697	0.0
4	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0
5	86.0	21.11111111	92.0	3.549	0.8053864	6.6994	4.81924	10.57635	773.92	0.0
6	49.0	22.85445769	92.0	3.226	0.732086933	6.8317	13.67975	10.3176	530.41	0.0
7	89.0	22.7	77.0	4.69	0.890787333	6.964	5.589865	12.9361	1256.083	0.0
8	76.0	23.8	118.0	6.47	1.883201333	4.311	13.25132	5.1042	280.694	0.0
9	73.0	22.0	97.0	3.35	0.801543333	4.47	10.358725	6.28445	136.855	0.0
10	75.0	23.0	83.0	4.952	1.013839467	17.127	11.57899	7.0913	318.302	0.0
11	34.0	21.47	78.0	3.469	0.6674356	14.57	13.11	6.92	354.6	0.0
12	29.0	23.01	82.0	5.663	1.145436133	35.59	26.72	4.58	174.8	0.0
13	25.0	22.86	82.0	4.09	0.827270667	20.45	23.67	5.14	313.73	0.0
14	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
15	38.0	23.34	75.0	5.782	1.06967	15.26	17.95	9.35	165.02	0.0
16	44.0	20.76	86.0	7.553	1.6	14.09	20.32	7.64	63.61	0.0
17	47.0	22.03	84.0	2.869	0.59	26.65	38.04	3.32	191.72	0.0
18	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
19	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
20	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	64.0	22.22222222	98.0	5.7	1.37788	12.1905	4.783985	13.91245	395.976	1.0
2	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
3	68.0	21.08281329	102.0	6.2	1.55992	9.6994	8.574655	13.74244	448.799	1.0
4	42.0	21.35991456	93.0	2.999	0.6879706	19.0826	8.462915	17.37615	321.919	1.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	50.0	38.57875854	106.0	6.703	1.752611067	46.6401	4.667645	11.78388	887.16	0.0
2	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
3	38.0	22.4996371	95.0	5.261	1.232827667	8.438	4.77192	15.73606	199.055	1.0
4	81.0	31.64036818	100.0	9.669	2.38502	38.8066	10.636525	29.5583	426.175	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	73.0	30.3005366865040 6	97.0	10.8415536479106 79	2.67014887958481 45	48.0836074608048 7	6.47461438367389	17.7227909095595 56	829.324908589366 7	0.0
2	82.0	28.1808678773576 13	96.0	4.68239483669355	1.12175306866207 2	21.9284833853802 82	7.72047928452810 7	23.1064479138403 9	237.868242054045 6	0.0
3	75.0	26.4412454619050 2	92.0	12.4227800289642 8	2.85615490632579 85	32.5011737731394 7	9.74991887634802 6	8.17635746790495 7	366.581854486369 8	0.0
4	39.0	27.1992231324848 46	87.0	4.07563424920180 7	0.87607100967411 68	13.1284844433025 04	9.82279774139153	9.43402593762632 5	533.305893678691 4	0.0
5	64.0	25.0569756969103 92	88.0	3.90502605968246 46	0.84900531800226 73	20.6967776354481 82	6.63981391168544	13.0994037947298 86	756.646924313616 1	0.0
6	60.0	29.9506206375511 7	91.0	13.3278620483579 43	2.87814034694891 33	28.3970096781224 05	5.7299755449089	16.0952223961935 74	420.390913679616 74	0.0
7	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
8	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
9	71.0	25.51020408	112.0	10.395	2.871792	19.0653	5.4861	42.7447	799.898	1.0
10	49.0	21.36752137	78.0	2.64	0.507936	6.3339	3.886145	22.94254	737.672	1.0

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B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	47.0	28.0594112320380	85.0	4.1759287400654	0.88219525306701	31.3474786632241	8.02510113743152	15.0192128007118	743.503722082655	0.0
		9			13	77	3	32	4	
2	74.0	32.4338325597531	99.0	5.27531690821209	1.29702824452884	59.8034338966112	7.15223809501530	59.4061487053342	253.639268066151	0.0
		2		05	68	24	2	4	52	
3	27.0	35.0401629903473	86.0	8.29444678569295	1.77682232333353	65.4938148483638	5.97889508752050	20.4203992157892	355.806333927084	0.0
		76		1	47	8	6	9	15	
4	28.0	35.3261771248416	86.0	8.32442268362056	1.78155529941887	65.9509998504799	5.00459219360258	20.8672135949939	361.131200074123	0.0
		3		3	84	5	2	4	9	
5	75.0	30.48	152.0	7.01	2.628282667	50.53	10.06	11.73	99.45	1.0
6	60.0	26.34929208	103.0	5.138	1.305394533	24.2998	2.19428	20.2535	378.996	0.0
7	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
8	72.0	23.62	105.0	4.42	1.14478	21.78	17.86	4.82	195.94	1.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	64.0	36.4459118440637	101.0	14.6612498186850	3.66076774609106	71.9359783704494	7.25601843663992	21.2800277983691	867.158926390004	0.0
		9		02			7	86	8	
2	57.0	30.7320064250788	82.0	13.8041930162917	2.87142433500518	27.5639684775496	6.40180306255057	12.6898539488626	453.247059131485	0.0
		5		59	02	15	5	91		
3	69.0	24.2028792338865	90.0	5.01880084392726	1.11854981355950	19.7030703942013	8.20852468810428	11.8184293975236	340.099905490974	0.0
		1		1	67	76		14	04	
4	35.0	29.6716476924786	86.0	4.40460847039285	0.93796462823846	19.5150121756258	8.29156966378997	8.01715680336984	546.648016269611	0.0
		43		3	83	77	1			
5	74.0	29.5931395559452	82.0	5.15433450499931	1.05335105848316	26.2735031703163	6.79439858620053	7.16228185586336	686.882687759322	0.0
		57		1	58	17	3	8	8	
6	70.0	22.9884070968652	82.0	4.88062083927033	0.99839036681167	17.4021658365483	12.5802036266519	6.92971977223690	317.923408906199	0.0
		52		7	03	4	2	2	54	
7	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
8	48.0	28.125	90.0	2.54	0.56388	15.5325	10.22231	16.11032	1698.44	1.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0

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B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	74.0	28.65013774	88.0	3.012	0.6538048	31.1233	7.65222	18.35574	572.401	1.0
2	65.0	32.05	97.0	5.73	1.370998	61.48	22.54	10.33	314.05	1.0
3	76.0	27.2	94.0	14.07	3.262364	35.891	9.34663	8.4156	377.227	0.0
4	65.0	29.666548	85.0	14.649	3.071407	26.5166	7.28287	19.46324	1698.44	1.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	69.0	30.7580934131646 73	91.0	9.49732572635563 5	2.12484935209673 6	54.3876869638945 3	7.93099788535930 1	23.0356788071556 7	227.155825441092 45	0.0
2	84.0	22.6856754874193	90.0	3.90383210970431 2	0.86273053175221 4	10.9454604716045 34	5.31462990633003 9	10.0844792824098 86	759.328379182159 4	0.0
3	35.0	30.27681661	84.0	4.376	0.9067072	39.2134	9.048185	16.43706	733.797	0.0
4	69.0	32.5	93.0	5.43	1.245642	15.145	11.78796	11.78796	270.142	0.0
5	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
6	49.0	29.77777778	70.0	8.396	1.449709333	51.3387	10.73174	20.76801	602.486	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	45.0	37.03560819	83.0	6.76	1.383997333	39.9802	4.617125	8.70448	586.173	0.0
2	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
3	46.0	22.21	86.0	36.94	7.836205333	10.16	9.76	5.68	312.0	1.0
4	85.0	27.68877813	196.0	51.814	25.05034187	70.8824	7.901685	55.2153	1078.359	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	78.0	25.3	60.0	3.508	0.519184	6.633	10.567295	4.6638	209.749	0.0
2	52.0	30.8012487	87.0	30.212	6.4834952	29.2739	6.26854	24.24591	764.667	1.0
3	45.0	20.26	92.0	3.44	0.780650667	7.65	16.67	7.84	193.87	1.0
4	59.0	28.67262608	77.0	3.188	0.605507467	17.022	16.44048	31.6904	910.489	1.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
2	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0
3	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
4	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0